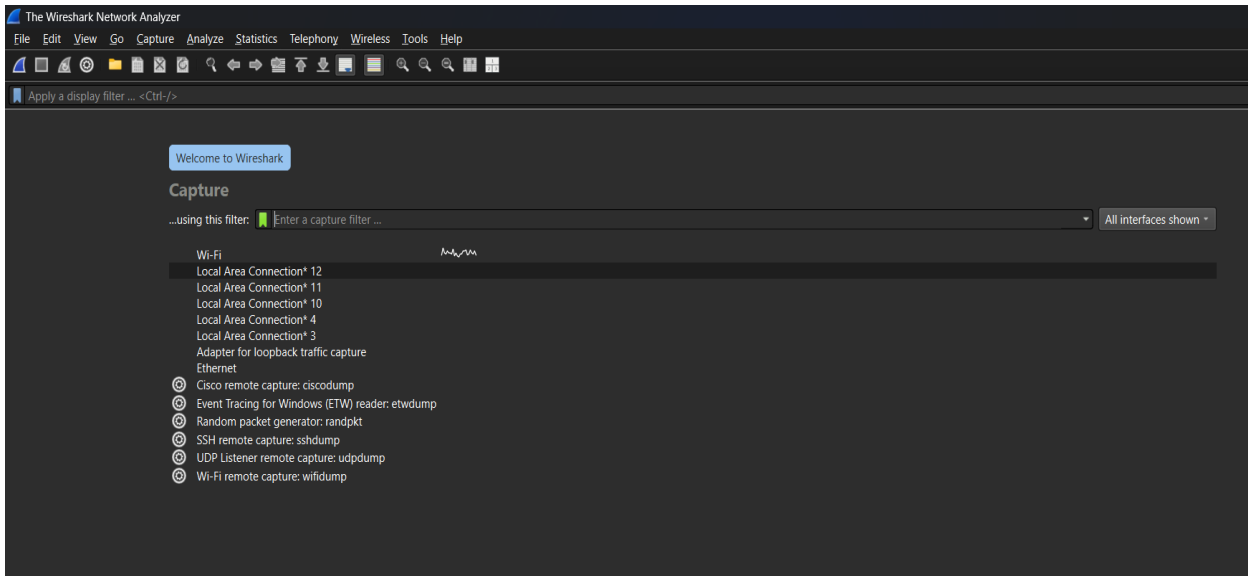


ASSIGNMENT 1

1. To capture live network packets using Wireshark and identify basic network protocols such as TCP, UDP, IP, and DNS.

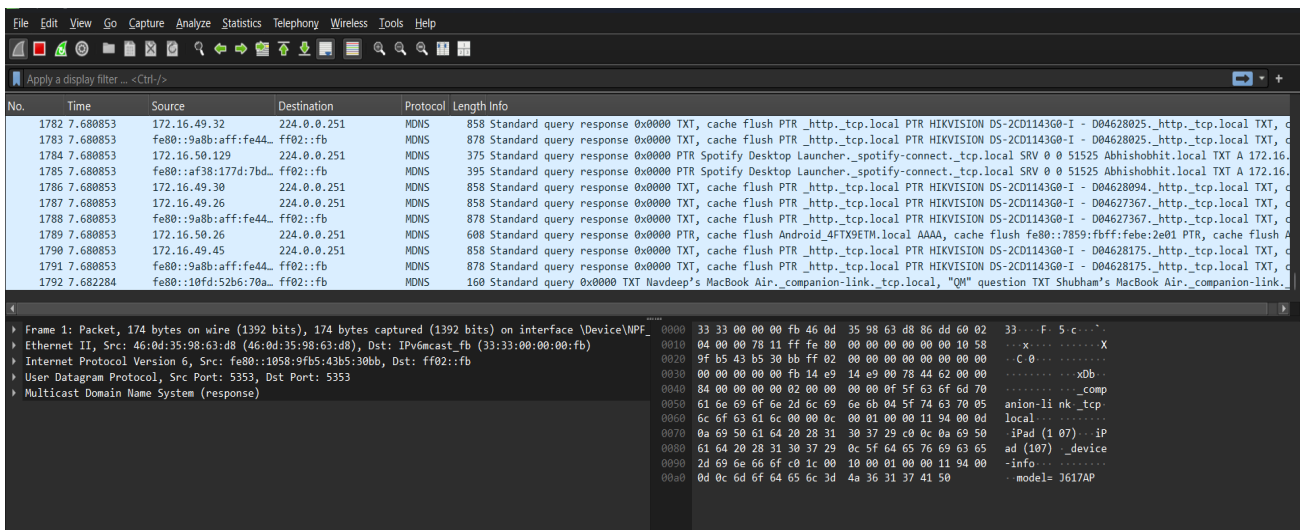
Step 1: Launch Wireshark

- Open Wireshark
- You will see a list of **network interfaces**
 - Wi-Fi
 - Ethernet



Step 2: Start Packet Capture

- Double-click on **Wi-Fi** (or active interface)
- Packet capture will start immediately



Step 3: Generate Network Traffic

Perform any of the following:

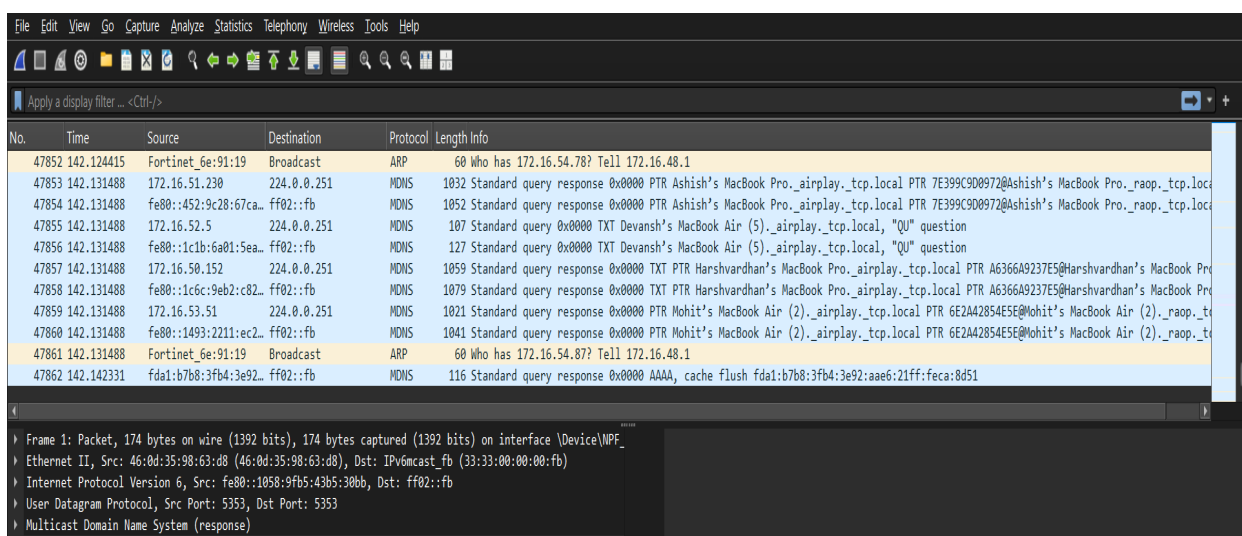
- Open a web browser
- Visit any website (e.g., google.com)
- Open Command Prompt and run:
- ping google.com

```
harsh — -zsh — 100x35

[(base) harsh@Harshs-MacBook-Air ~ % ping google.com
PING google.com (142.250.182.238): 56 data bytes
64 bytes from 142.250.182.238: icmp_seq=0 ttl=118 time=16.323 ms
64 bytes from 142.250.182.238: icmp_seq=1 ttl=118 time=59.844 ms
64 bytes from 142.250.182.238: icmp_seq=2 ttl=118 time=15.463 ms
64 bytes from 142.250.182.238: icmp_seq=3 ttl=118 time=14.400 ms
64 bytes from 142.250.182.238: icmp_seq=4 ttl=118 time=14.795 ms
64 bytes from 142.250.182.238: icmp_seq=5 ttl=118 time=51.621 ms
64 bytes from 142.250.182.238: icmp_seq=6 ttl=118 time=95.490 ms
64 bytes from 142.250.182.238: icmp_seq=7 ttl=118 time=14.350 ms
64 bytes from 142.250.182.238: icmp_seq=8 ttl=118 time=14.314 ms
64 bytes from 142.250.182.238: icmp_seq=9 ttl=118 time=16.518 ms
64 bytes from 142.250.182.238: icmp_seq=10 ttl=118 time=15.811 ms
64 bytes from 142.250.182.238: icmp_seq=11 ttl=118 time=18.380 ms
^C
--- google.com ping statistics ---
12 packets transmitted, 12 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 14.314/28.942/95.490/25.026 ms
(base) harsh@Harshs-MacBook-Air ~ %
```

Step 4: Stop Capture

- Click the red Stop button
- Captured packets will remain on the screen



Step 5: Apply Basic Filters

Use the **Display Filter** bar:

Filter	Purpose
ip	Show IP packets
tcp	Show TCP packets
udp	Show UDP packets
dns	Show DNS packets
icmp	Show ping packets

tcp					
No.	Time	Source	Destination	Protocol	Length Info
6892	13.725480	172.16.41.135	13.204.106.34	TCP	55 60059 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
6896	13.728040	13.204.106.34	172.16.41.135	TCP	66 443 → 60059 [ACK] Seq=1 Ack=2 Win=103 Len=0 SLE=1 SRE=2
7705	14.957805	172.16.41.135	74.125.200.188	TCP	55 58464 → 5228 [ACK] Seq=1 Ack=1 Win=253 Len=1
7746	15.038603	74.125.200.188	172.16.41.135	TCP	66 5228 → 58464 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2
8973	17.342393	172.16.41.135	52.111.224.7	TCP	66 56569 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
8976	17.345076	52.111.224.7	172.16.41.135	TCP	66 443 → 56569 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM WS=256
8977	17.345148	172.16.41.135	52.111.224.7	TCP	54 56569 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
8979	17.346649	172.16.41.135	52.111.224.7	TLShv1.2	250 Client Hello (SNI=messaging.engagement.office.com)
8983	17.349960	52.111.224.7	172.16.41.135	TCP	60 443 → 56569 [ACK] Seq=1 Ack=197 Win=15872 Len=0
9196	17.701531	52.111.224.7	172.16.41.135	TLShv1.2	1514 Server Hello
9197	17.701531	52.111.224.7	172.16.41.135	TCP	1490 443 → 56569 [PSH, ACK] Seq=1461 Ack=197 Win=15872 Len=1436 [TCP PDU reassembled in 9203]
9198	17.701630	172.16.41.135	52.111.224.7	TCP	54 56569 → 443 [ACK] Seq=197 Ack=2897 Win=65280 Len=0
9200	17.702894	52.111.224.7	172.16.41.135	TCP	1514 443 → 56569 [ACK] Seq=2897 Ack=197 Win=15872 Len=1460 [TCP PDU reassembled in 9203]
9201	17.702894	52.111.224.7	172.16.41.135	TCP	1490 443 → 56569 [PSH, ACK] Seq=4357 Ack=197 Win=15872 Len=1436 [TCP PDU reassembled in 9203]
9203	17.702894	52.111.224.7	172.16.41.135	TLShv1.2	1502 Certificate
9204	17.702894	52.111.224.7	172.16.41.135	TLShv1.2	1514 Certificate Status
9205	17.702894	52.111.224.7	172.16.41.135	TLShv1.2	362 Server Key Exchange, Server Hello Done
9206	17.702961	172.16.41.135	52.111.224.7	TCP	54 56569 → 443 [ACK] Seq=197 Ack=5793 Win=65280 Len=0
9207	17.703110	172.16.41.135	52.111.224.7	TCP	54 56569 → 443 [ACK] Seq=197 Ack=9009 Win=65280 Len=0
9208	17.706135	172.16.41.135	52.111.224.7	TLShv1.2	180 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
Frame 6892: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on interface \Device\NPF_{000004d5-906e-910f-0000-000000000000} on 0 (04:d5:90:6e:91:0f)					
Ethernet II, Src: Intel_0a:fe:60 (98:43:fa:0a:fe:60), Dst: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)					
Internet Protocol Version 4, Src: 172.16.41.135, Dst: 13.204.106.34					
Transmission Control Protocol, Src Port: 60059, Dst Port: 443, Seq: 1, Ack: 1, Len: 1					

udp					
No.	Time	Source	Destination	Protocol	Length Info
6913	13.792808	fe80::cbc:18e:982f::	ff02::fb	MDNS	480 Standard query response 0x0000 PTR Gautam's MacBook Pro._companion-link._tcp.local TXT TXT, cache flush AAAA, cache flush fe80::1489::
6914	13.792952	172.16.37.209	224.0.0.251	MDNS	456 Standard query response 0x0000 PTR Gautam's MacBook Pro._companion-link._tcp.local TXT TXT, cache flush AAAA, cache flush fe80::1489::
6917	13.803743	172.16.41.135	173.194.154.38	UDP	75 51085 → 443 Len=33
6918	13.808004	172.16.40.74	224.0.0.251	MDNS	476 Standard query response 0x0000 PTR Bose Tech 11.1._companion-link._tcp.local TXT TXT, cache flush AAAA, cache flush fe80::1489::
6919	13.808486	fe80::1489:2c1f:251::	ff02::fb	MDNS	500 Standard query response 0x0000 PTR Bose Tech 11.1._companion-link._tcp.local TXT TXT, cache flush AAAA, cache flush fe80::1489::
6921	13.815118	172.16.38.188	224.0.0.251	MDNS	100 Standard query 0x0000 PTR _rdlink._tcp.local, "QM" question PTR _companion-link._tcp.local, "QM" question
6922	13.815246	fe80::148f:fb02:f3f::	ff02::fb	MDNS	124 Standard query 0x0000 PTR _rdlink._tcp.local, "QM" question PTR _companion-link._tcp.local, "QM" question
6923	13.820218	fe80::180c:b040:79d::	ff02::fb	MDNS	781 Standard query 0x0000 PTR _rdlink._tcp.local, "QM" question PTR _companion-link._tcp.local, "QM" question PTR Hardik's MacBook...
6924	13.820368	172.16.39.97	224.0.0.251	MDNS	757 Standard query 0x0000 PTR _rdlink._tcp.local, "QM" question PTR _companion-link._tcp.local, "QM" question PTR Hardik's MacBook...
6925	13.834393	fe80::855:e380:a211::	ff02::fb	MDNS	268 Standard query response 0x0000 TXT, cache flush NSEC, cache flush Air._companion-link._tcp.local
6926	13.834598	172.16.39.147	224.0.0.251	MDNS	244 Standard query response 0x0000 TXT, cache flush NSEC, cache flush Air._companion-link._tcp.local
6927	13.862722	fe80::2628:fdff:fec::	ff02::fb	MDNS	424 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G31130507._http._tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU ...
6928	13.863016	172.16.33.73	224.0.0.251	MDNS	400 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G31130507._http._tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU ...
6929	13.864682	fe80::2628:fdff:fef::	ff02::fb	MDNS	424 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459854._http._tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU ...
6930	13.865402	172.16.33.94	224.0.0.251	MDNS	400 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459854._http._tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU ...
6931	13.866687	173.194.154.38	172.16.41.135	UDP	1296 443 → 51085 Len=1250
6933	13.872134	172.16.41.135	173.194.154.38	UDP	75 51085 → 443 Len=33
6934	13.874120	fe80::2628:fdff:fef::	ff02::fb	MDNS	453 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459853._http._tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU ...
6935	13.874120	fe80::7305:21e8:a91::	ff02::fb	MDNS	112 Standard query 0x0000 ANY Aditya(59)._dosvc._tcp.local, "QU" question
6936	13.874359	172.16.33.95	224.0.0.251	MDNS	429 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459853._http._tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU ...
Frame 6891: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface \Device\NPF_{000004d5-906e-910f-0000-000000000000} on 0 (04:d5:90:6e:91:0f)					
Ethernet II, Src: Intel_0a:fe:60 (98:43:fa:0a:fe:60), Dst: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)					
Internet Protocol Version 4, Src: 172.16.41.135, Dst: 173.194.154.38					
User Datagram Protocol, Src Port: 51085, Dst Port: 443					
Data (33 bytes)					

No.	dns dnsserver	Source	Destination	Protocol	Length	Info
3100 6.291883	172.16.41.135	172.31.1.6	DNS	70	Standard query 0xe0dd A google.com	
3101 6.291474	172.16.41.135	172.31.1.6	DNS	70	Standard query 0xd228 AAAA google.com	
3103 6.293499	172.31.1.6	172.16.41.135	DNS	86	Standard query response 0xe0dd A google.com A 172.217.24.238	
3105 6.319569	172.31.1.6	172.16.41.135	DNS	90	Standard query response 0xd228 AAAA google.com AAAA 2404:6800:4002:82a::200e	
8967 17.321777	172.16.41.135	172.31.1.6	DNS	91	Standard query 0xd109 A messaging.engagement.office.com	
8968 17.321967	172.16.41.135	172.31.1.6	DNS	91	Standard query 0x8464 AAAA messaging.engagement.office.com	
8969 17.323584	172.31.1.6	172.16.41.135	DNS	182	Standard query response 0xd109 A messaging.engagement.office.com CNAME prod-campaignaggregator.omexexternalfb.office.net.akad_	
8972 17.337200	172.31.1.6	172.16.41.135	DNS	194	Standard query response 0x8464 AAAA messaging.engagement.office.com CNAME prod-campaignaggregator.omexexternalfb.office.net.a_	
9440 18.080831	172.16.41.135	172.31.1.6	DNS	86	Standard query 0x853b A recent.svc.cloud.microsoft	
9441 18.080967	172.16.41.135	172.31.1.6	DNS	86	Standard query 0xe3c2 AAAA recent.svc.cloud.microsoft	
9444 18.084769	172.31.1.6	172.16.41.135	DNS	368	Standard query response 0x853b A recent.svc.cloud.microsoft CNAME prod.ocws1.live.com.akadns.net CNAME asia2.ocws1.live.com.ak_	
9445 18.110026	172.16.41.135	8.8.8.8	DNS	86	Standard query 0xe3c2 AAAA recent.svc.cloud.microsoft	
9449 18.139246	172.31.1.6	172.16.41.135	DNS	464	Standard query response 0xe3c2 AAAA recent.svc.cloud.microsoft CNAME prod.ocws1.live.com.akadns.net CNAME asia2.ocws1.live.com_	
9492 18.257844	172.16.41.135	172.31.1.6	DNS	80	Standard query 0x32c2 A substrate.office.com	
9493 18.258247	172.16.41.135	172.31.1.6	DNS	80	Standard query 0x4e9f AAAA substrate.office.com	
9495 18.270105	172.31.1.6	172.16.41.135	DNS	266	Standard query response 0x32c2 A substrate.office.com CNAME outlook.office365.com CNAME ooc-g2.tm-4.office.com A 40.104.79.2 A_	
9500 18.283716	172.31.1.6	172.16.41.135	DNS	302	Standard query response 0x4e9f AAAA substrate.office.com CNAME outlook.office365.com CNAME ooc-g2.tm-4.office.com CNAME outloo_	
9501 18.283834	172.16.41.135	8.8.8.8	DNS	80	Standard query 0x4e9f AAAA substrate.office.com	
9515 18.300042	8.8.8.8	172.16.41.135	DNS	302	Standard query response 0x4e9f AAAA substrate.office.com CNAME outlook.office365.com CNAME ooc-g2.tm-4.office.com CNAME outloo_	
10048 19.497391	172.16.41.135	172.31.1.6	DNS	86	Standard query 0x5279 A editor.svc.cloud.microsoft	

Frame 3105: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface \Device\NPF_{...} ...
 Ethernet II, Src: Fortinet_6e:91:0f (04:d5:90:6e:91:0f), Dst: Intel_0a:fe:60 (98:43:fa:0a:fe:60) ...
 Internet Protocol Version 4, Src: 172.31.1.6, Dst: 172.16.41.135 ...
 User Datagram Protocol, Src Port: 53, Dst Port: 50387 ...
 Domain Name System (response) ...

No.	icmp icmpv6	Source	Destination	Protocol	Length	Info
3111 6.3004199	172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request	id=0x0001, seq=1/256, ttl=128 (reply in 3127)
3127 6.380559	172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply	id=0x0001, seq=1/256, ttl=118 (request in 3111)
3513 7.342610	172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request	id=0x0001, seq=2/512, ttl=128 (reply in 3518)
3518 7.364507	172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply	id=0x0001, seq=2/512, ttl=118 (request in 3513)
4290 8.357826	172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request	id=0x0001, seq=3/768, ttl=128 (reply in 4323)
4323 8.394766	172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply	id=0x0001, seq=3/768, ttl=118 (request in 4290)
4906 9.374616	172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request	id=0x0001, seq=4/1024, ttl=128 (reply in 4911)
4911 9.402421	172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply	id=0x0001, seq=4/1024, ttl=118 (request in 4906)

Step 6: Analyze a Packet

- Click on any packet
- Observe:
 - Ethernet header
 - IP header
 - TCP/UDP header
 - Payload data

Wireshark - Packet 4906 - Wi-Fi	
Frame 4906: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{30E8F03D-A225-4923-BF48-48BA90E0F267}, id 0 Ethernet II, Src: Intel_0a:fe:60 (98:43:fa:0a:fe:60), Dst: Fortinet_6e:91:0f (04:d5:90:6e:91:0f) Destination: Fortinet_6e:91:0f (04:d5:90:6e:91:0f) Source: Intel_0a:fe:60 (98:43:fa:0a:fe:60) Type: IPv4 (0x0800) [Stream index: 4]	
Internet Protocol Version 4, Src: 172.16.41.135, Dst: 172.217.24.238 0100 = Version: 4 0101 = Header Length: 20 bytes (5) Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT) Total Length: 60 Identification: 0x2c90 (11408) 000. = Flags: 0x0 ...0 0000 0000 0000 = Fragment Offset: 0 Time to Live: 128 Protocol: ICMP (1) Header Checksum: 0x0000 [validation disabled] [Header checksum status: Unverified] Source Address: 172.16.41.135 Destination Address: 172.217.24.238 [Stream index: 177]	
Internet Control Message Protocol	
0000 04 d5 90 6e 91 0f 98 43 fa 0a fe 60 08 00 45 00 ... n . C E . 0010 00 3c 2c 90 00 00 80 01 00 00 ac 10 29 87 ac d9 < ,) . . . 0020 18 ee 08 00 4d 57 00 01 00 04 61 62 63 64 65 66 ... PMW . . . abcdef 0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv 0040 77 61 62 63 64 65 66 67 68 69 wabcedfg hi	